

JavaScript Programming, Problem Solving & DSA Training Plan

Duration: 12 Sessions × 3 Hours

Goal: Build strong JavaScript fundamentals, problem solving skills, algorithms, data structures, OOP, asynchronous programming, Git/GitHub, and database fundamentals.

Topics Covered During Training

Programming Fundamentals

Variables, Data Types, Input/Output, Operators, Debugging

Control Flow

if/else, nested conditions, switch

Loops

for, while, do-while, nested loops

Functions

Parameters, Return Values, Scope, Arrow Functions

Arrays

Traversal, Search, Insert, Delete, Update

Strings

Methods, Palindrome Check, Reverse String, Word Count

Objects

Objects, Nested Objects, Arrays of Objects

Object-Oriented Programming

Classes, Objects, Constructors, Encapsulation, Inheritance, Polymorphism, Static Methods

Problem Solving

Problem Analysis, Pseudocode, Flowcharts, Dry Runs, Debugging

Algorithms

Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort, Big O Basics

Data Structures

Arrays, Stack, Queue, Set, Map

Asynchronous JavaScript

Callbacks, Promises, Async/Await, Error Handling, Fetch API

Database Fundamentals

Tables, Records, Relationships, Primary Keys, Foreign Keys, Basic SQL

Git & GitHub

Init, Add, Commit, Push, Pull, Branches, Merge, Pull Requests

Problem Solving Platforms

LeetCode: Arrays, Strings, Searching, Sorting, Stack, Queue

Codeforces: Beginner Div 4 and Div 3 problems

Mini Projects

- Calculator
- Student Management System
- Library Management System
- Center Management System
- API Data Fetcher

Self-Study After Training

Advanced JavaScript: Destructuring, Spread Operator, Rest Operator, Closures, Hoisting, Modules, Event Loop

Advanced Problem Solving: Recursion, Two Pointers, Sliding Window, Prefix Sum, Binary Search Variations

Advanced Data Structures: Linked List, Doubly Linked List, Tree, BST, Heap, Trie, Graph

Advanced Algorithms: Merge Sort, Quick Sort, DFS, BFS, Greedy Algorithms, Dynamic Programming

Databases: MySQL, PostgreSQL, ERD Design, Normalization, Indexing

Software Engineering: SOLID, Design Patterns, Clean Code, Refactoring

Git Advanced: Rebase, Cherry Pick, Stash, Git Flow

Career Preparation: LeetCode 150, Codeforces Journey, Technical Interviews, CV Building, GitHub Portfolio